Nested YOSO MPC:
Near Optimal Resilience Without an MPC Setup

Ittai Abraham, Eli Chouatt, Ivan Damgård, Yossi Gilad, Gilad Stern, and **Sophia Yakoubov**

Results

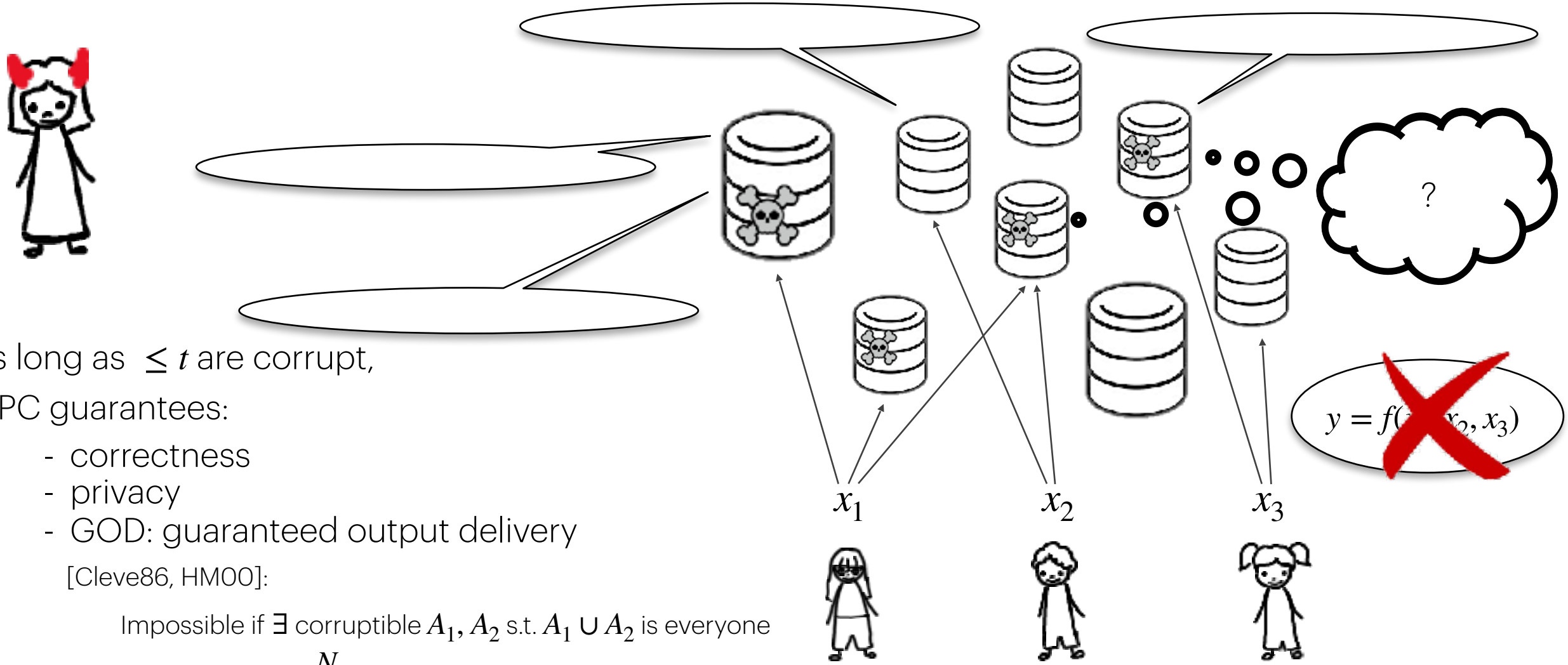
Efficient YOSO MPC for $N\left(\frac{1}{2}+\epsilon\right)$ honest parties without trusted setup

parties

Lower bound: YOSO MPC impossible if $< N\left(\frac{1}{2}+\frac{\epsilon}{B}\right)$ parties are honest

broadcast complexity

Multi-Party Computation (MPC)



As long as $\leq t$ are corrupt,

MPC guarantees:

- correctness
- privacy
- GOD: guaranteed output delivery

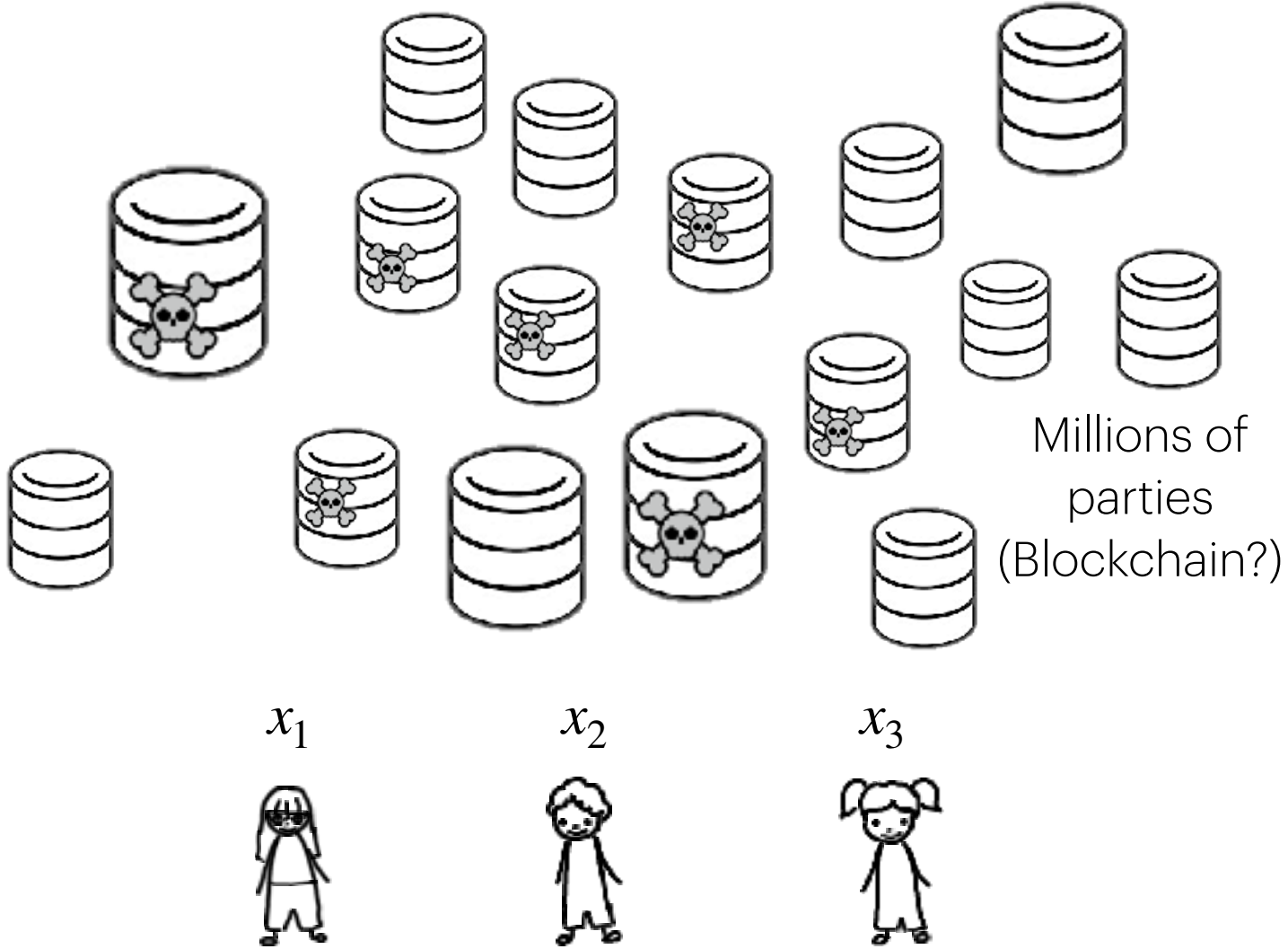
[Cleve86, HM00]:

Impossible if ∃ corruptible A_1, A_2 s.t. $A_1 \cup A_2$ is everyone

i.e. we need. $t < \frac{N}{2}$

Multi-Party Computation (MPC)





Security goals

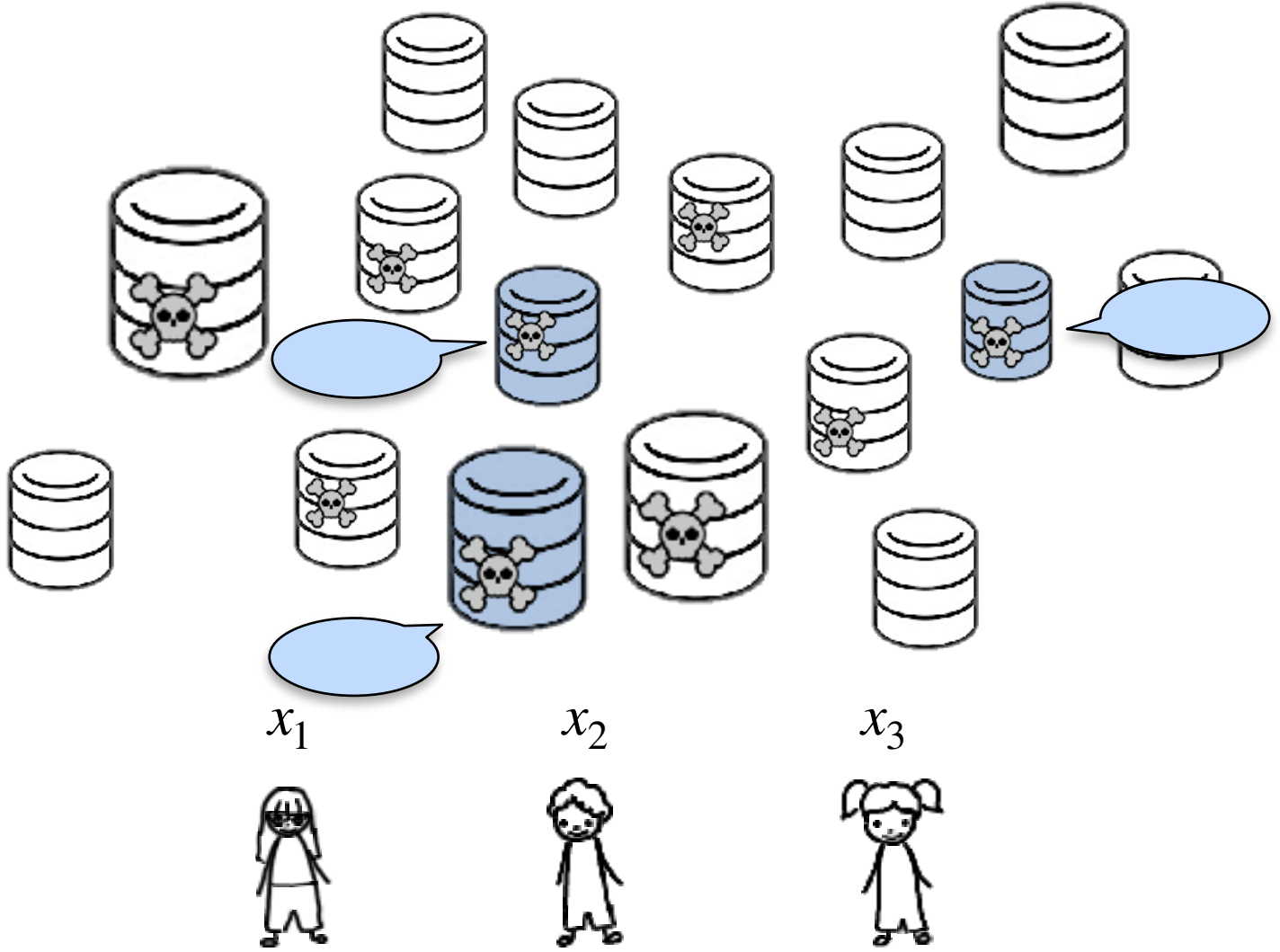
- correctness
- privacy
- GOD

Efficiency goals

- Communication sublinear in N

Multi-Party Computation (MPC)





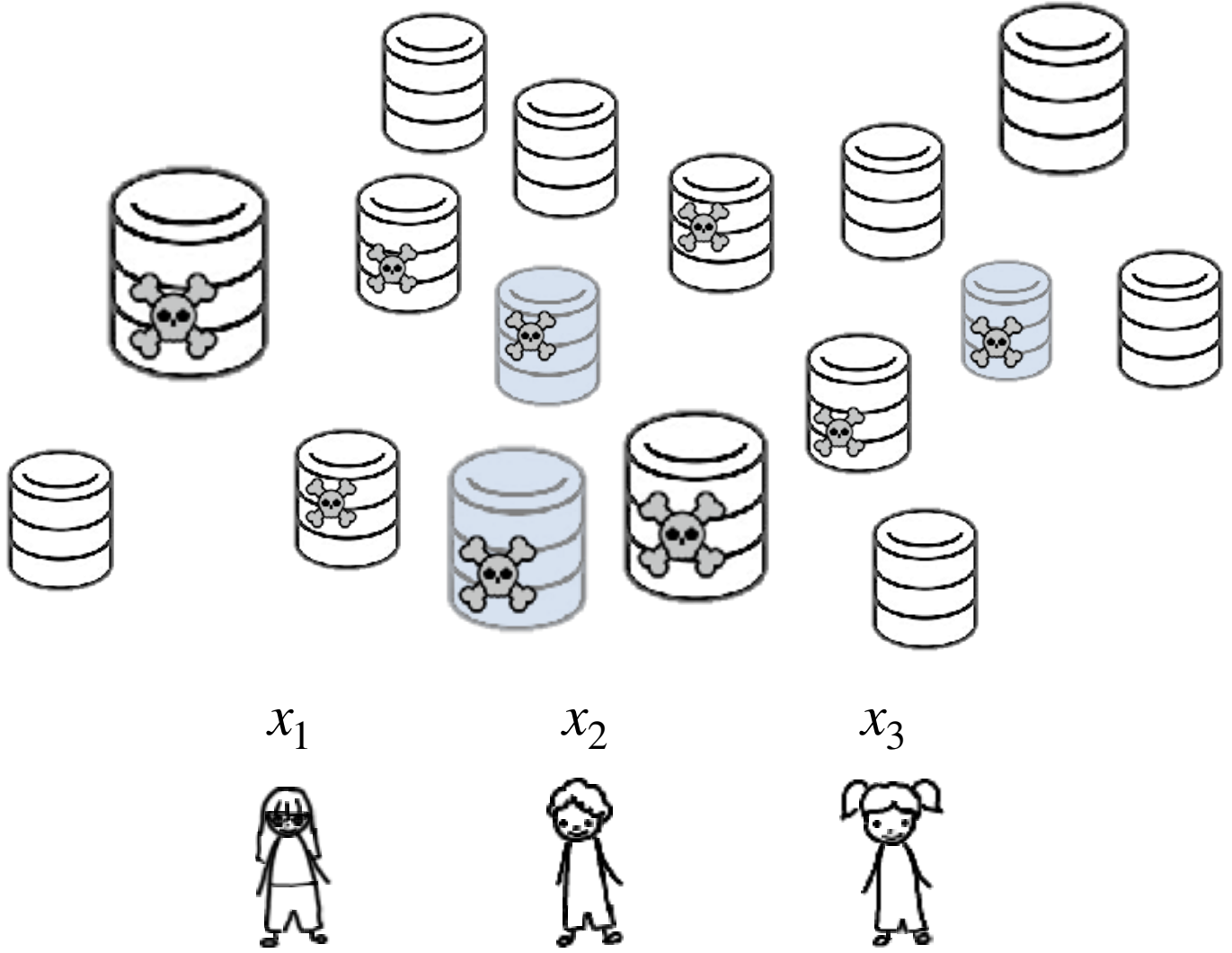
Security goals
- correctness
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Efficiency goals
- Communication sublinear in N

even with *adaptive corruptions*

You Only Speak Once (YOSO) MPC





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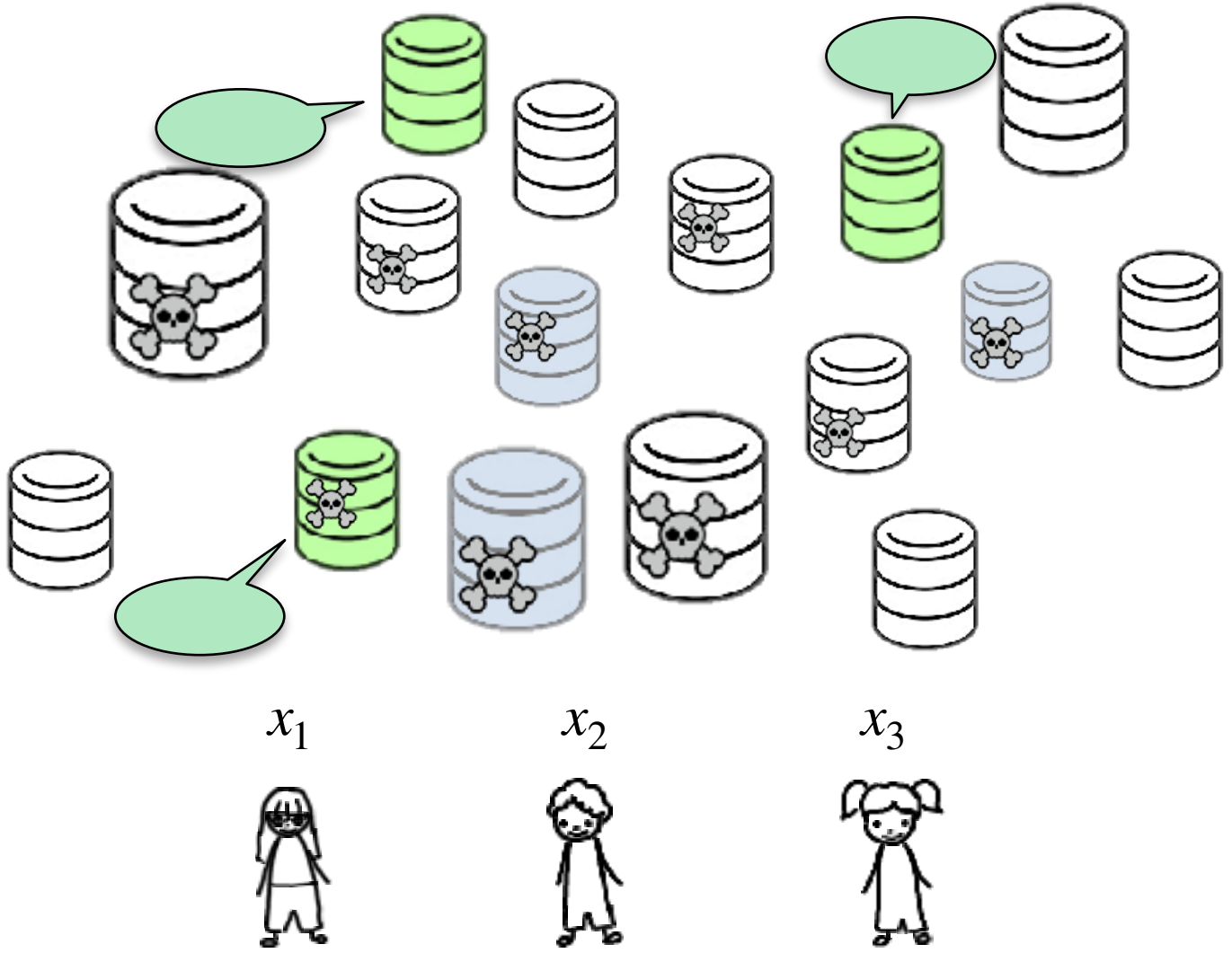
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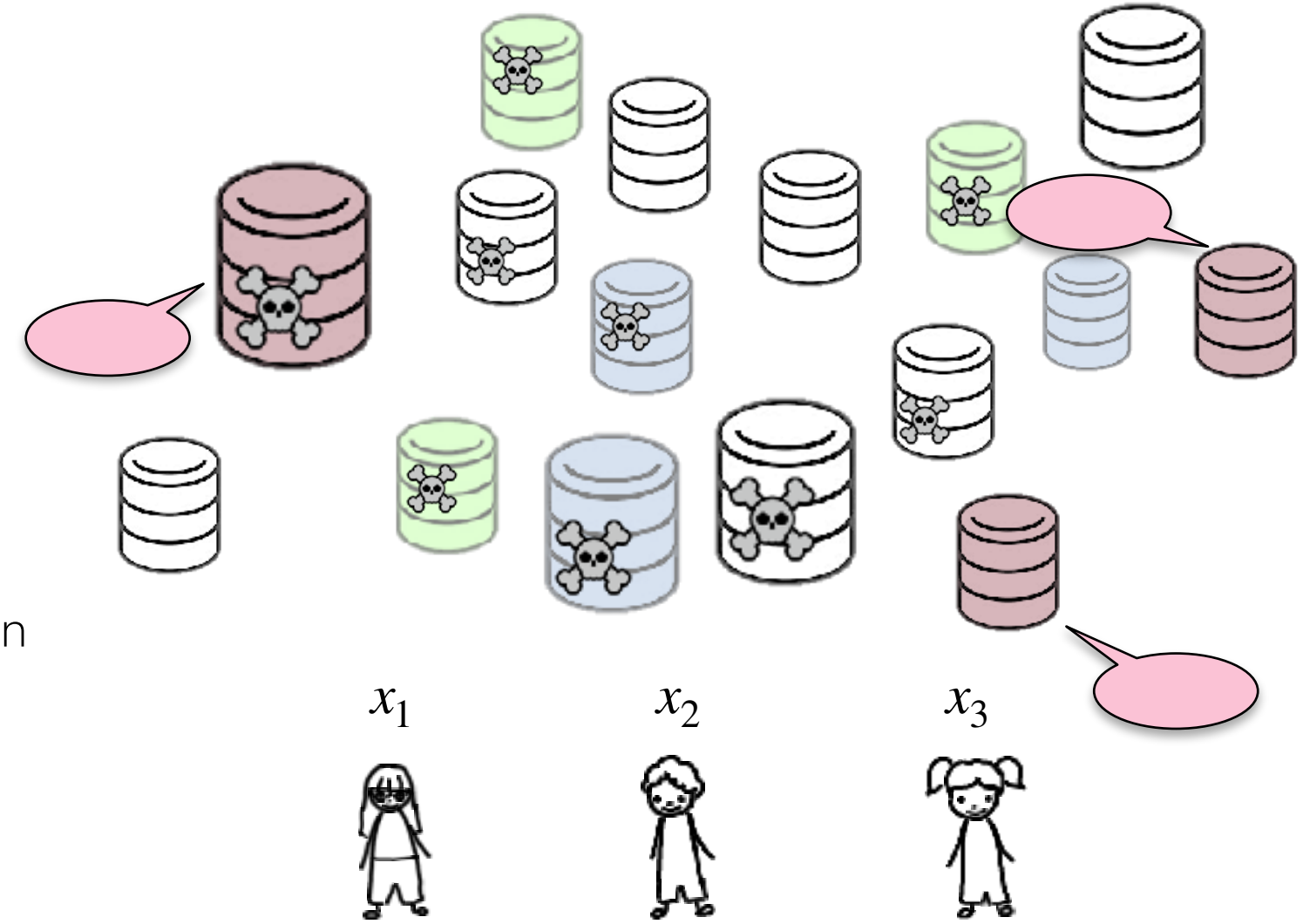
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You Only Speak Once (YOSO) MPC





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Results

Efficient YOSO MPC for $N\left(\frac{1}{2}+\epsilon\right)$ honest parties without trusted setup

- Role Assignment
- Computation

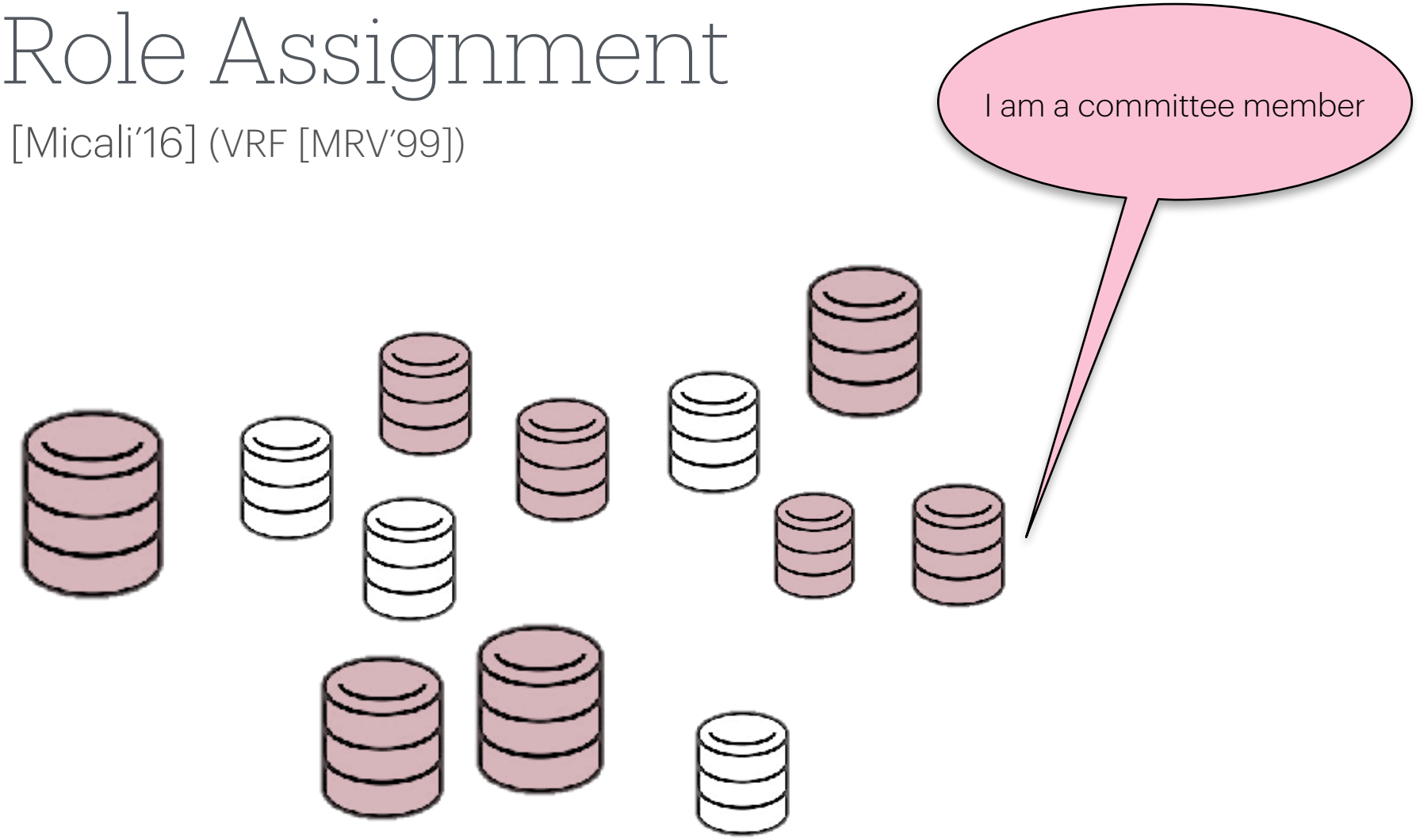
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Role Assignment



Role Assignment

[Micali'16] (VRF [MRV'99])

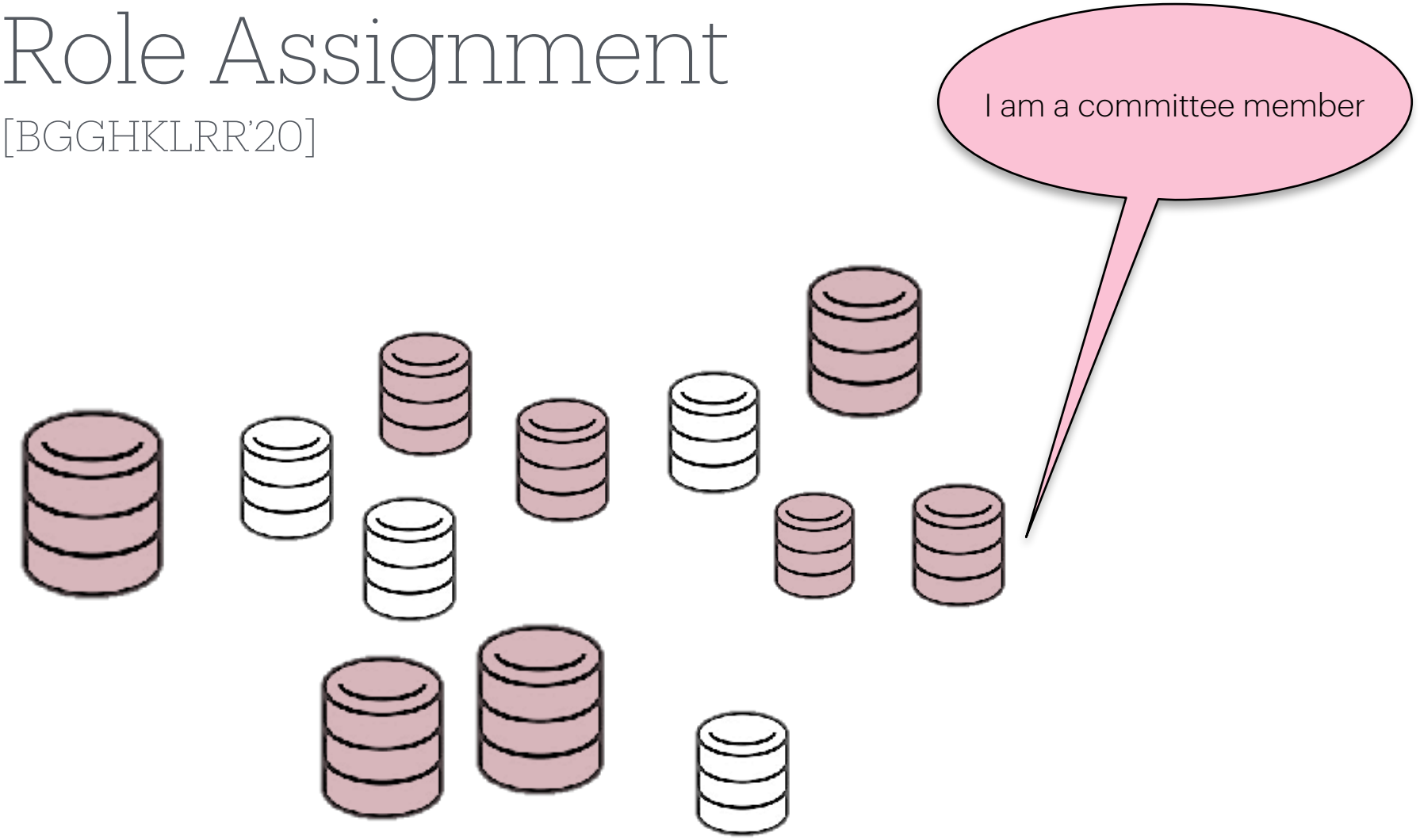


sortition: winning determined by local VRF evaluation
- provable
- a-priori unpredictable

challenge: sending secret messages to committee members

Role Assignment

[BGGHKLRR'20]



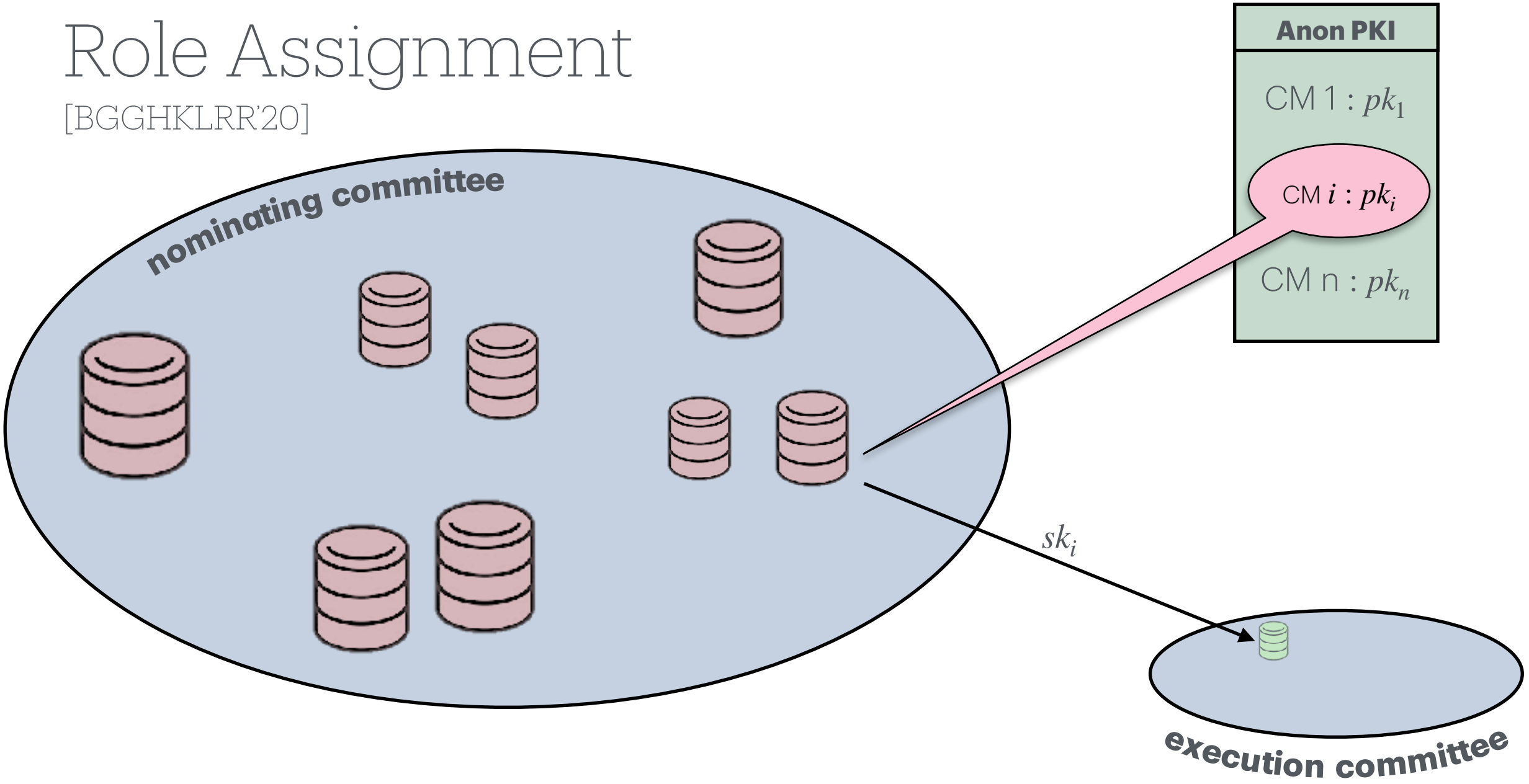
Anon PKI
CM 1 : pk_1
⋮
CM n : pk_n

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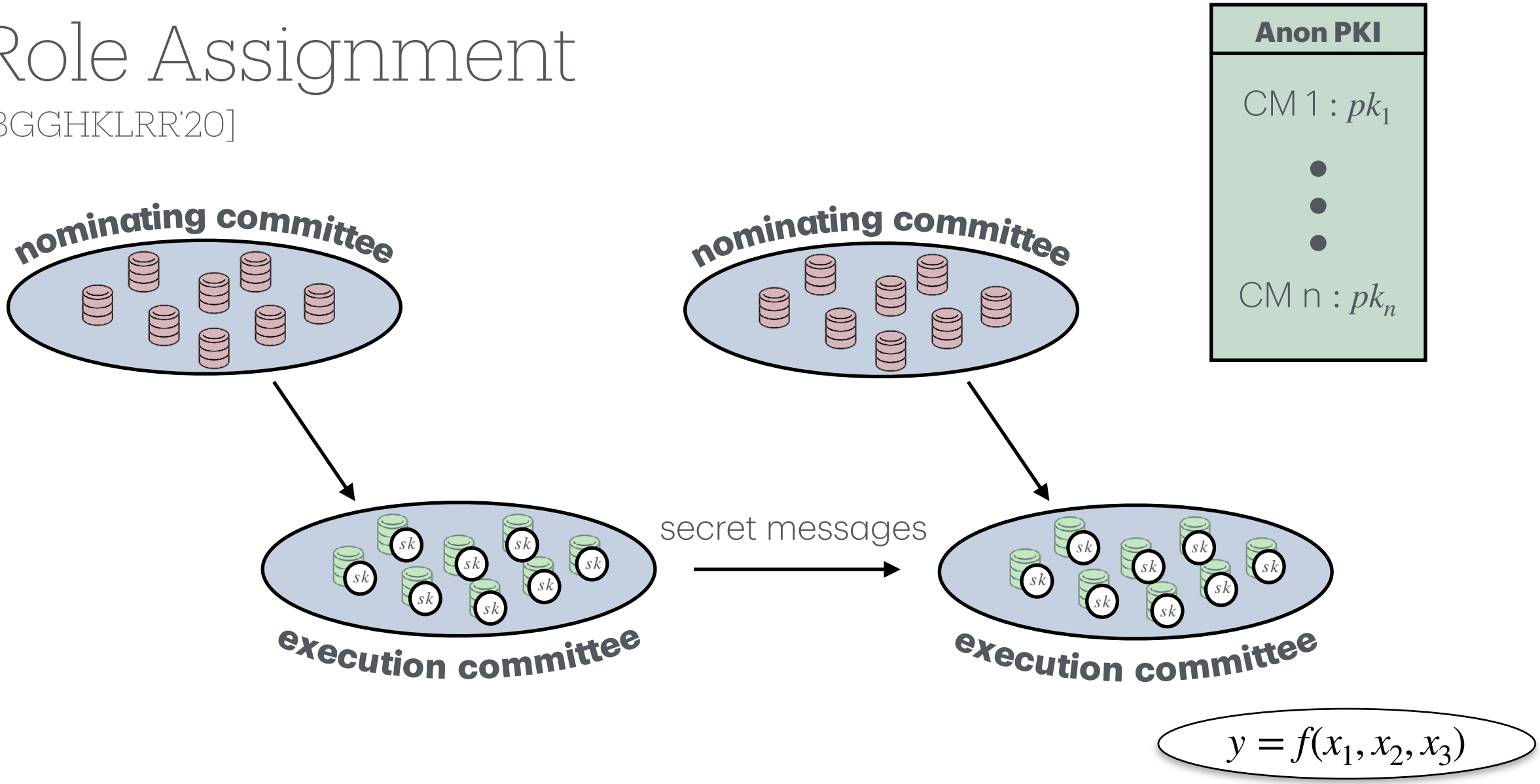
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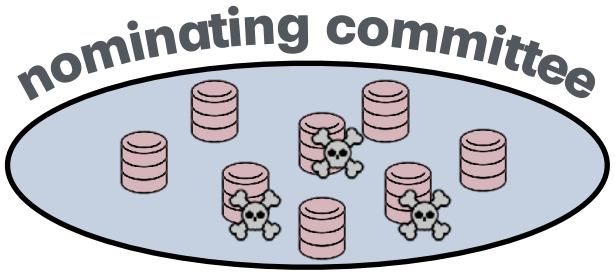
Role Assignment

[BGGHKLRR'20]

Recall [Cleve86, HM00]:

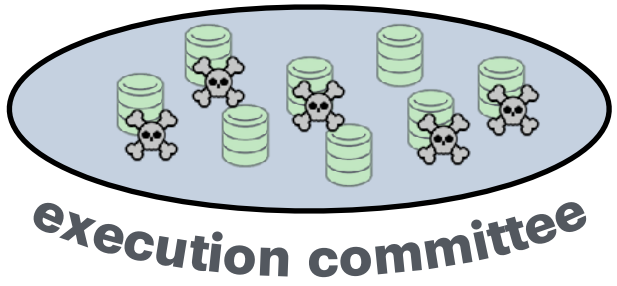
cannot get guaranteed output delivery

if ∃ corruptible A_1, A_2 s.t. $A_1 \cup A_2$ is everyone



how many can we tolerate?

we need to corrupt $< 2/3N$ to get the adversarial committees.



Role Assignment Approaches

	~Optimal Threshold $t = \left(\frac{1}{2} - \epsilon\right)N$
[BGGHKLRR'20] [GHKMBRY'21]	NO

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[CDGK22]	YES

Role Assignment Approaches

	~Optimal Threshold $t = \left(\frac{1}{2} - \epsilon\right)N$	Communication sublinear in N?
[BGGHKLRR'20] [GHKMBRY'21]	NO	YES
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[CDGK22]	YES	NO
[GHMNY21,DKLN25]	YES	YES

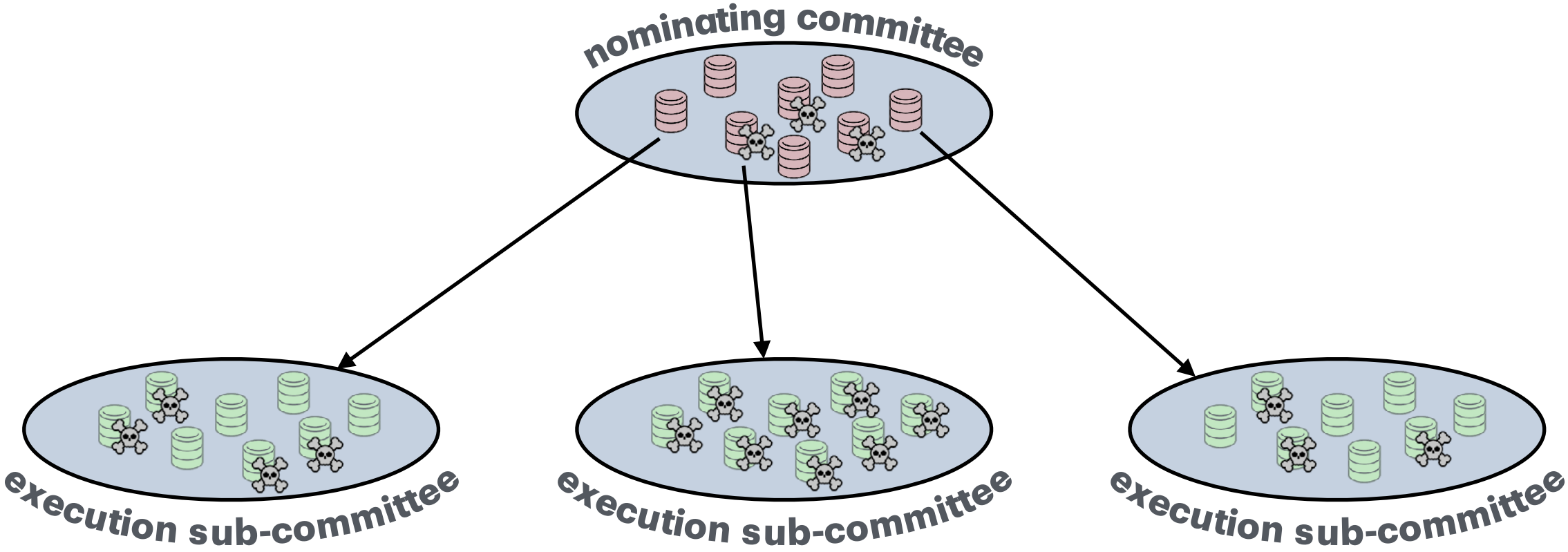
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[BGGHKLRR'20] [GHKMBRY'21]	NO	YES	YES
[CDGK22]	YES	NO	YES
[GHMNY21,DKLN25]	YES	YES	NO
Nested YOSO [ACDGSY26]	YES	YES	YES

Our Work: Nested Committees



n *separate* n-party execution sub-committees!

Guarantee: **most** of these will be **mostly** honest

$\nexists$ corruptible A_1, A_2 s.t. $A_1 \cup A_2$ is everyone!

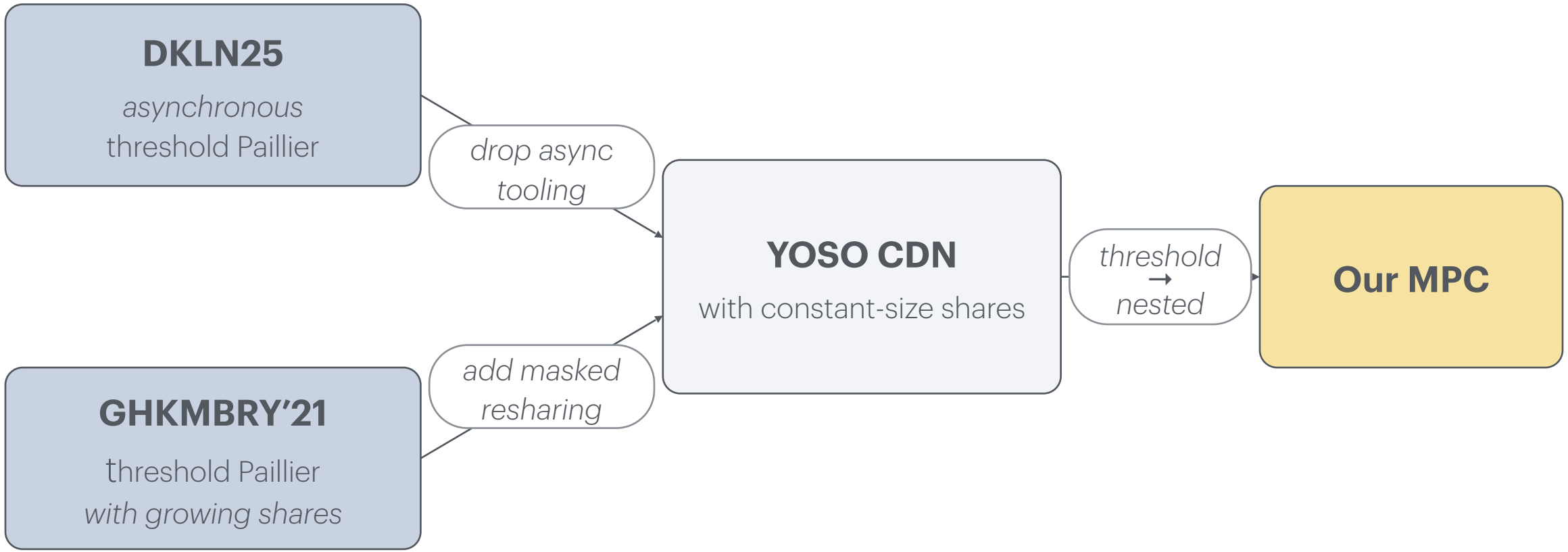
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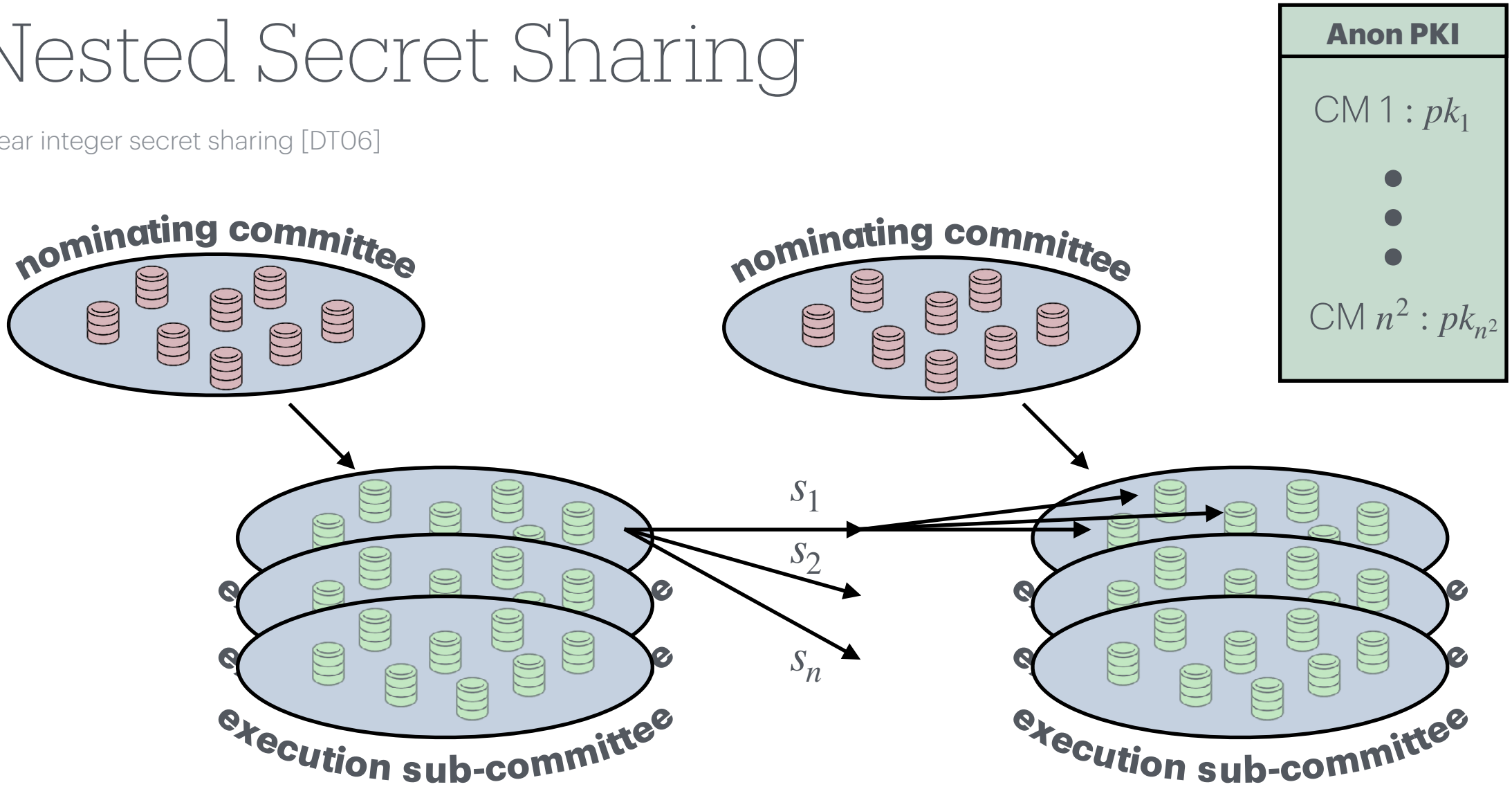
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Our MPC Protocol



Nested Secret Sharing

linear integer secret sharing [DT06]



- *sk* is threshold-shared into $s_1, \ldots, s_n$
- each s_i re-shared inside sub-committee i

Reconstruct from majority of most sub-committees

Future Work

- Smaller committees?
- Fewer rounds?

